



Tomographic and Biomechanical Stability of the Non-operated Eye in Asymmetric Keratoconus with Unilateral Intracorneal Ring Segment Implantation

João A. Ambrósio · Júlio Brissos · Gonçalo C. Tardão · Alexandre Neto ·

Louise Esporcatte · Renato Ambrósio Jr.

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ABSTRACT

Introduction: This study aimed to assess the tomographic and biomechanical progression of non-operated eyes in patients with asymmetric keratoconus who underwent unilateral intracorneal ring segment (ICRS) implantation.

Methods: This retrospective study included 33 patients (66 eyes) with asymmetric keratoconus who underwent unilateral ICRS implantation. The fellow eye did not meet clinical or tomographic criteria for surgical intervention at the

time of the initial surgery. Tomographic and biomechanical data from the non-operated fellow eye were compared between baseline and the final follow-up (minimum 12 months). Scheimpflug-based corneal tomography (Pentacam®) assessed keratometry, curvature, corneal irregularity indices, and thickness-based metrics. Biomechanics parameters were evaluated via dynamic corneal response analysis (Corvis ST®). Disease progression was defined as the presence of at least two concordant changes among the following: Kmax increase > 1.0 D, thinnest corneal thickness decrease > 20 µm, BAD-D increase > 0.42, and worsening of biomechanical E-stage.

João A. Ambrósio and Júlio Brissos contributed equally to this work and share first authorship.

J. A. Ambrósio
Department of Ophthalmology, Unidade Local de Saúde Entre Douro e Vouga, Santa Maria da Feira, Portugal

J. Brissos
Department of Ophthalmology, Unidade Local de Saúde de São José, Lisbon, Portugal

G. C. Tardão
Department of Ophthalmology, Unidade Local de Saúde Almada-Seixal, Almada, Portugal

A. Neto · R. Ambrósio Jr.
Department of Ophthalmology, Federal University the State of Rio de Janeiro (UNIRIO), Rio de Janeiro, Brazil

L. Esporcatte · R. Ambrósio Jr.
Department of Ophthalmology, Federal University of São Paulo, São Paulo, Brazil

L. Esporcatte · R. Ambrósio Jr.
Rio de Janeiro Corneal Tomography and Biomechanics Study Group, Rio de Janeiro, Brazil

L. Esporcatte · R. Ambrósio Jr.
Instituto de Olhos Renato Ambrósio, Rio de Janeiro, Brazil

R. Ambrósio Jr. (✉)
RioVision Hospital, Rua Prudente de Moraes, 1276, Rio de Janeiro, RJ 22420-042, Brazil
e-mail: dr.renatoambrosio@gmail.com

R. Ambrósio Jr.
Brazilian Study Group of Artificial Intelligence and Corneal Analysis-BRAIN, Rio de Janeiro and Maceió, Brazil

Results: Over a median follow-up of 31 months, non-operated fellow eyes showed no clinically meaningful changes in visual, tomographic, or biomechanical parameters, suggesting no structural progression under conservative, behavior-based management, including counseling to avoid eye rubbing, management of atopic disease, and oral riboflavin supplementation. Disease progression was observed in only two eyes (6.1%), which were subsequently referred for corneal cross-linking (CXL). ICRS-treated eyes demonstrated significant improvements in visual acuity and corneal regularity.

Conclusion: In this paired-eye study, most non-operated fellow eyes in asymmetric keratoconus remained stable under conservative, behavior-based management combined with regular multimodal monitoring. These results support a behavior-first approach, using corneal tomography and biomechanics to detect objective progression and guide timely intervention. Conservative management in asymmetric keratoconus should be understood as an active, follow-up-dependent strategy rather than the absence of treatment. Surgical interventions should remain reserved for objectively progressing cases, recognizing that a small but clinically relevant proportion of fellow eyes still progress and require CXL.

Keywords: Keratoconus; Intracorneal ring segment; Behavioral management; Fellow eye stability; Corneal tomography; Corneal biomechanics

Key Summary Points

In this paired-eye cohort of clinically asymmetric keratoconus treated unilaterally with intracorneal ring segments (ICRS), non-operated fellow eyes remained stable over a median follow-up of 31 months, with no clinically meaningful deterioration in visual, tomographic, or biomechanical parameters.

A structured behavior-first strategy, including patient and family education, eye rubbing avoidance, allergy control, and oral riboflavin supplementation, was effective in maintaining corneal stability in most non-operated fellow eyes under conservative management.

Objective tomographic progression occurred in only two eyes ($\approx 6\%$), which required timely corneal cross-linking, indicating that progression may still occur in a small subset despite conservative management.

These findings support observation with clinical and behavioral measures as a safe approach only when combined with continuous, regular multimodal monitoring (tomography and biomechanics), reinforcing the need for sustained surveillance rather than reduced follow-up.

INTRODUCTION

Keratoconus is a bilateral, progressive ectatic disorder characterized by localized stromal thinning and conical protrusion of the cornea, leading to irregular astigmatism and variable visual impairment. The disease typically manifests during adolescence or early adulthood and may progress until the fourth decade of life before stabilizing [1]. Reported prevalence varies widely worldwide, reflecting substantial geographic and methodological heterogeneity [2].

Although keratoconus is inherently a bilateral condition, it often manifests asymmetrically, encompassing a continuum from manifest disease to subclinical and forme fruste stages [3, 4]. Although universally accepted diagnostic criteria for these early phenotypes are lacking, advances in corneal imaging have improved their detection [4, 5]. The combination of Scheimpflug-based tomography and dynamic biomechanical analysis, particularly when integrated with artificial intelligence (AI)-driven algorithms, has enhanced the ability to detect ectatic disease at an early preclinical stage and to identify cases of very asymmetric ectasia with greater precision [4–8].

Chronic mechanical trauma from eye rubbing is recognized as a significant environmental factor associated with both the onset and progression of keratoconus [9, 10]. Repetitive rubbing can cause microstructural disruption and localized weakening of the corneal stroma, leading to focal thinning and steepening. The frequent observation of greater severity in one eye supports the hypothesis that asymmetric keratoconus may result from preferential rubbing of the more affected eye [11]. Early management should prioritize behavioral and medical measures to halt disease progression, including strict avoidance of eye rubbing, control of ocular allergy with topical antiallergic and lubricating agents, and visual rehabilitation with spectacles or contact lenses when appropriate [12, 13]. Public awareness initiatives such as Violet June highlight the role of patient education and avoidance of eye rubbing in keratoconus [14]. As an adjunct to conservative care, emerging evidence suggests that high-dose oral riboflavin supplementation may promote corneal stiffening and contribute to keratoconus stabilization, although clinical evidence remains limited [15, 16].

A paradigm shift in keratoconus management has moved from corneal transplantation toward minimally invasive treatments that preserve tissue and improve optical quality [12, 17]. Corneal cross-linking (CXL) remains the standard approach to halt ectatic progression by increasing stromal stiffness through photo-induced collagen cross-linking [18, 19]. Intracorneal ring segments (ICRS) involve the implantation of synthetic material, such as polymethyl methacrylate (PMMA) arcs, within the corneal stroma to flatten the cone and regularize corneal curvature [20]. This procedure improves visual acuity [21, 22], reduces higher-order aberrations [23], and facilitates contact lens fitting [24]. In selected cases, ICRS implantation may also delay or prevent the need for keratoplasty [25]. However, long-term studies have reported variable outcomes regarding its influence on disease progression [26–28].

Although keratoconus is inherently bilateral, the rate and pattern of progression may differ substantially between eyes, even within the same patient. This asymmetry reflects

the combined influence of local biomechanical susceptibility, environmental factors, and behavioral habits [9–11]. In patients undergoing unilateral ICRS implantation, the fellow, non-operated eye serves as a natural control to assess the intrinsic course of keratoconus and to understand bilateral disease behavior after unilateral surgical intervention. This study aimed to evaluate tomographic and biomechanical changes over time in these non-operated eyes to characterize their evolution in the absence of surgical treatment.

METHODS

This retrospective study included patients with a diagnosis of keratoconus who underwent ICRS implantation in only one eye. Cases were identified from institutional clinical records. Only patients whose fellow eye did not meet the clinical or tomographic criteria for any surgical intervention, including ICRS implantation or CXL, at the time of the initial surgery were included. The contralateral, non-operated eyes served as controls. Eligible patients were required to have a minimum follow-up of 12 months. Patients with any underlying conditions other than keratoconus or a history of any previous ocular surgery were excluded from the study.

Ethics

This study, conducted at the Instituto de Olhos Renato Ambrósio, was approved by the Comitê de Ética em Pesquisa da UNIFESP—Universidade Federal de São Paulo—Hospital São Paulo (approval number 4.050.934; 27784919.0.0000.5505). All procedures were conducted in accordance with the ethical standards of the institutional and national research committees and with the principles of the Helsinki Declaration of 1964 and its later amendments. Given the retrospective design, informed consent was waived, and only anonymized, non-identifiable clinical data were used in compliance with institutional guidelines and data protection standards.

Analysis

Tomographic and biomechanical data from the non-operated fellow eye were compared between baseline (the preoperative evaluation of the contralateral eye before ICRS implantation) and the final follow-up visit with a minimum follow-up of 12 months. Follow-up visits were not protocolized and reflected routine clinical practice. For analytical consistency, the final follow-up visit was defined as the last available assessment, including both tomography and biomechanical evaluation. All patients underwent comprehensive ophthalmologic assessment, including Scheimpflug-based corneal tomography (Pentacam®, Oculus, Wetzlar, Germany) and dynamic corneal response analysis (Corvis ST®, Oculus, Wetzlar, Germany). Changes in uncorrected visual acuity (UCVA) and best-corrected visual acuity (BCVA) were also analyzed.

All patients received routine disease counseling, including education on keratoconus and explicit counseling to avoid eye rubbing. Ocular allergy and atopic disease were managed as clinically indicated, including avoidance counseling and topical therapy (antihistamine/mast-cell stabilizer agents and lubricants), with referral to immunoallergology when appropriate. All patients also received oral riboflavin supplementation (400 mg/day) as part of routine care. Given the retrospective, real-world nature of the cohort, adherence to these recommendations was not formally quantified and therefore was not included as a variable in the statistical analysis.

Tomographic Analysis

Tomographic evaluation was divided into three main categories: (1) Keratometry and curvature, assessed by maximum keratometry (Kmax), mean keratometry (Kmean), anterior radius of curvature (ARC), and posterior radius of curvature (PRC); (2) Corneal irregularity indices, including the central keratoconus index (CKI), index of surface variance (ISV), index of vertical asymmetry (IVA), index of height asymmetry (IHA), and index of height decentration (IHD);

and (3) Pachymetry and thickness-based metrics, evaluated by thinnest corneal thickness (TCT), pachymetric progression index (PPI), and Ambrósio relational thickness (ARTmax and ARTavg). The Belin–Ambrósio deviation index (BAD-D) was also analyzed as a global tomographic indicator of ectasia.

Biomechanical Analysis

Biomechanical assessment included the Corvis biomechanical index (CBI), its linearized form, the Corvis biomechanical factor (CBiF), and the corresponding biomechanical staging classification (cE-Stage) [29]. Additional parameters analyzed were the deformation amplitude ratio (DA-Ratio), integrated radius (IR), stiffness parameter at first appplanation (SP-A1), stress–strain index (SSI), and the tomographic and biomechanical index (TBI).

Progression Analysis

Tomographic progression was defined according to the following criteria: (1) an increase in Kmax > 1.0 D within 12 months, (2) a reduction in TCT > 20 µm over the same period [29], or (3) an increase in the BAD-D > 0.42 [30]. Biomechanical progression was defined as an increase in the Corvis biomechanical factor (cE-Stage) from E0 to E4. Overall disease progression was considered when at least two of the aforementioned criteria were met.

Statistical Analysis

All statistical analyses were performed using SPSS software version 30.0 (IBM Corp., Armonk, NY, USA). The normality of continuous variables was assessed using the Shapiro–Wilk test. Longitudinal changes between baseline and follow-up were analyzed using a two-way repeated-measures analysis of variance (ANOVA), with time (baseline vs follow-up) as the within-subjects factor and surgery status (operated vs non-operated eyes) as the between-subjects factor. Main effects of time, between-group differences, and time × surgery interaction effects were analyzed. When significant effects were detected,

Sidak-adjusted pairwise comparisons were used to identify specific within- and between-group differences. Effect sizes for ANOVA were expressed as partial eta-squared (η^2) and interpreted as small (0.01), moderate (0.06), or large (0.14). A two-tailed p value <0.05 was considered statistically significant.

RESULTS

Participants and Follow-up

A total of 66 eyes (33 operated and 33 non-operated) from 33 patients (25 men [75.8%] and 8 women [24.2%]) were included. The mean age was 23.5 ± 2.1 years (range 18–38), and the median follow-up was 31.3 months (Q1=17.5, Q3=46.7; range 12–126). ICRS implantation was performed in 17 left eyes and 16 right eyes.

As shown in Table 1, baseline and last follow-up parameters were analyzed in surgically treated and fellow non-surgically treated eyes to assess longitudinal changes and treatment effects.

Visual Acuity

UCVA improved overall ($F(1,64)=21.18$, $p<0.001$, $\eta^2=0.26$), with a significant time $\times$ surgery interaction ($F(1,64)=19.11$, $p<0.001$, $\eta^2=0.24$); Sidak tests showed significant gains only in ICRS eyes, with no change in non-operated eyes. BCVA also showed a time effect ($F(1,64)=30.84$, $p<0.001$, $\eta^2=0.33$) and interaction ($F(1,64)=14.44$, $p<0.001$, $\eta^2=0.19$); visual acuity gains were restricted to the surgery group.

Keratometry and Curvature

Kmax demonstrated a strong time effect ($F(1,64)=24.51$, $p<0.001$, $\eta^2=0.28$) and a larger interaction ($F(1,64)=45.49$, $p<0.001$, $\eta^2=0.42$), indicating pronounced flattening in ICRS eyes and stability in non-operated eyes. Mean K changed over time ($F(1,64)=6.61$, $p=0.013$, $\eta^2=0.10$) with interaction ($F(1,64)=11.18$, $p=0.001$, $\eta^2=0.15$), again indicating progressive flattening after ICRS implantation and no

change in the fellow eyes. The ARC decreased significantly after surgery, reflecting corneal flattening in the ICRS-treated eyes, while remaining stable in the non-operated eyes (time effect: $F(1,64)=36.61$, $p<0.001$, $\eta^2=0.39$; time $\times$ surgery interaction: $F(1,64)=39.47$, $p<0.001$, $\eta^2=0.41$). The PRC did not change overall ($F(1,64)=0.71$, $p=0.403$, $\eta^2=0.01$) but showed a significant time $\times$ surgery interaction ($F(1,64)=12.19$, $p<0.001$, $\eta^2=0.17$), indicating that the curvature changes were limited to the operated eyes.

Corneal Irregularity Indices

ISV and IVA displayed time effects (ISV: $F=8.29$, $p=0.006$, $\eta^2=0.12$; IVA: $F=8.32$, $p=0.005$, $\eta^2=0.12$) and time $\times$ surgery interactions (ISV: $F=20.35$, $p<0.001$, $\eta^2=0.25$; IVA: $F=17.13$, $p<0.001$, $\eta^2=0.22$). Sidak tests confirmed significant improvement after ICRS and relative stability in non-operated eyes. The KI decreased significantly over time in the ICRS-treated eyes, indicating improved corneal regularity, while remaining stable in the non-operated eyes (time effect: $F(1,64)=15.85$, $p<0.001$, $\eta^2=0.21$; time $\times$ surgery interaction: $F(1,64)=26.97$, $p<0.001$, $\eta^2=0.31$). The CKI also showed a significant time effect ($F(1,64)=7.49$, $p=0.008$, $\eta^2=0.11$) but no interaction, reflecting a mild overall reduction across both groups. The IHA remained unchanged. In contrast, the IHD improved significantly in the operated eyes but remained stable in the fellow eyes (time effect: $F(1,64)=17.23$, $p<0.001$, $\eta^2=0.22$; interaction: $F(1,64)=29.43$, $p<0.001$, $\eta^2=0.33$), again reflecting structural improvement after ICRS implantation.

Pachymetry and Thickness-Based Metrics

The PPI showed a time effect ($F=7.26$, $p=0.009$, $\eta^2=0.11$) and interaction ($F=4.19$, $p=0.045$, $\eta^2=0.06$), with greater attenuation of progression in ICRS eyes and no change in non-operated eyes. Minimum pachymetry remained stable (no significant effects). ART-Max and ART-Avg showed time effects (ART-Max: $F=21.20$, $p<0.001$, $\eta^2=0.26$; ART-Avg:

Table 1 Repeated-measures ANOVA with time-by-surgery interaction analysis

	Pre		Post		F-tests		Sidak Pre vs. post	
	NS	S	NS	S	Main-effects	Interaction effects	NS	S
UCVA (logMAR)	0.34 (0.07)	0.70 (0.06)	0.33 (0.06)	0.37 (0.05)	$F(1,64) = 21.18$, $p < 0.001, \eta^2 = 0.26$	$F(1,64) = 19.11$, $p < 0.001, \eta^2 = 0.24$	0.873	< 0.001
BCVA (logMAR)	0.09 (0.04)	0.36 (0.04)	0.05 (0.02)	0.16 (0.02)	$F(1,64) = 30.84$, $p < 0.001, \eta^2 = 0.33$	$F(1,64) = 14.44$, $p < 0.001, \eta^2 = 0.19$	0.227	< 0.001
Kmax (D)	48.30 (0.82)	55.71 (0.82)	48.77 (0.83)	52.64 (0.83)	$F(1,64) = 24.51$, $p < 0.001, \eta^2 = 0.28$	$F(1,64) = 45.49$, $p < 0.001, \eta^2 = 0.42$	0.209	< 0.001
Kmean (D)	43.65 (0.42)	46.46 (0.42)	43.77 (0.38)	45.55 (0.38)	$F(1,64) = 6.61$, $p = 0.013, \eta^2 = 0.10$	$F(1,64) = 11.18$, $p = 0.001, \eta^2 = 0.15$	0.587	< 0.001
ARC (mm)	7.45 (0.08)	6.76 (0.08)	7.44 (0.08)	7.15 (0.08)	$F(1,64) = 36.61$, $p < 0.001, \eta^2 = 0.387$	$F(1,64) = 39.47$, $p < 0.001, \eta^2 = 0.405$	0.870	< 0.001
PRC (mm)	5.82 (0.09)	5.03 (0.09)	5.75 (0.09)	5.14 (0.09)	$F(1,64) = 0.709$, $p = 0.403, \eta^2 = 0.012$	$F(1,64) = 12.19$, $p < 0.001, \eta^2 = 0.174$	0.066	0.003
TCT (μm)	488.33 (8.61)	456.18 (8.61)	484.76 (10.44)	456.24 (10.44)	$F(1,64) = 0.62$, $p = 0.434, \eta^2 = 0.01$	$F(1,64) = 0.66$, $p = 0.418, \eta^2 = 0.01$	0.262	0.995
PPI (Max)	7.66 (3.97)	2.94 (4.03)	7.75 (3.88)	3.66 (3.94)	$F(1,64) = 7.26$, $p = 0.009, \eta^2 = 0.11$	$F(1,64) = 4.19$, $p = 0.045, \eta^2 = 0.06$	0.646	0.001
PPI (Avg)	10.05 (8.75)	11.06 (8.89)	10.26 (8.23)	10.11 (8.36)	$F(1,64) = 0.37$, $p = 0.546, \eta^2 = 0.01$	$F(1,64) = 0.93$, $p = 0.339, \eta^2 = 0.02$	0.800	0.275
ART (Max)	270.16 (14.41)	166.97 (14.64)	243.31 (13.25)	147.77 (13.46)	$F(1,64) = 21.20$, $p < 0.001, \eta^2 = 0.26$	$F(1,64) = 0.59$, $p = 0.447, \eta^2 = 0.01$	< 0.001	< 0.001
ART (Avg)	363.87 (15.82)	235.19 (15.82)	342.29 (16.84)	219.00 (16.84)	$F(1,64) = 12.51$, $p = 0.001, \eta^2 = 0.17$	$F(1,64) = 0.25$, $p = 0.616, \eta^2 = 0.00$	0.006	0.036
BAD-D	4.11 (0.46)	8.35 (0.46)	4.46 (0.46)	7.82 (0.46)	$F(1,64) = 0.35$, $p = 0.559, \eta^2 = 0.01$	$F(1,64) = 8.76$, $p = 0.004, \eta^2 = 0.12$	0.098	0.015
ISV	48.16 (5.71)	89.36 (5.71)	51.71 (5.47)	73.29 (5.47)	$F(1,64) = 8.29$, $p = 0.006, \eta^2 = 0.12$	$F(1,64) = 20.35$, $p < 0.001, \eta^2 = 0.25$	0.253	< 0.001
IVA	0.55 (0.08)	1.02 (0.08)	0.59 (0.08)	0.80 (0.08)	$F(1,64) = 8.32$, $p = 0.005, \eta^2 = 0.12$	$F(1,64) = 17.13$, $p < 0.001, \eta^2 = 0.22$	0.379	< 0.001
IHA	24.90 (4.88)	35.37 (4.88)	25.02 (3.88)	28.00 (3.88)	$F(1,64) = 1.25$, $p = 0.267, \eta^2 = 0.02$	$F(1,64) = 1.33$, $p = 0.253, \eta^2 = 0.02$	0.980	0.113
IHD	0.067 (0.010)	0.14 (0.010)	0.07 (0.01)	0.097 (0.01)	$F(1,64) = 17.23$, $p < 0.001, \eta^2 = 0.223$	$F(1,64) = 29.43$, $p < 0.001, \eta^2 = 0.33$	0.371	< 0.001
KI	1.12 (0.02)	1.24 (0.02)	1.13 (0.02)	1.17 (0.02)	$F(1,64) = 15.85$, $p < 0.001, \eta^2 = 0.21$	$F(1,64) = 26.97$, $p < 0.001, \eta^2 = 0.31$	0.395	< 0.001

Table 1 continued

	Pre		Post		F-tests		Sidak Pre vs. post	
	NS	S	NS	S	Main-effects	Interaction effects	NS	S
CKI	1.01 (0.01)	1.06 (0.01)	1.02 (0.01)	1.07 (0.01)	$F(1,64)=7.49$, $p=0.008$, $\eta^2=0.11$	$F(1,64)=1.05$, $p=0.309$, $\eta^2=0.02$	0.231	0.010
DA-ratio	5.096 (0.123)	5.64 (0.12)	5.12 (0.17)	5.69 (0.17)	$F(1,64)=0.17$, $p=0.684$, $\eta^2=0.003$	$F(1,64)=0.03$, $p=0.872$, $\eta^2=0.00$	0.862	0.688
IR	10.195 (0.326)	11.91 (0.33)	10.17 (0.30)	11.65 (0.30)	$F(1,64)=0.57$, $p=0.453$, $\eta^2=0.009$	$F(1,64)=0.38$, $p=0.539$, $\eta^2=0.01$	0.923	0.335
SP-A1	77.489 (2.596)	60.97 (2.60)	76.76 (3.57)	65.57 (3.57)	$F(1,64)=0.62$, $p=0.435$, $\eta^2=0.010$	$F(1,64)=1.17$, $p=0.283$, $\eta^2=0.02$	0.834	0.191
SSI	0.852 (0.029)	0.72 (0.03)	0.82 (0.03)	0.76 (0.03)	$F(1,64)=0.022$, $p=0.881$, $\eta^2=0.000$	$F(1,64)=2.05$, $p=0.157$, $\eta^2=0.035$	0.372	0.264
CBI	0.807 (0.038)	0.94 (0.04)	0.79 (0.04)	0.92 (0.05)	$F(1,64)=1.09$, $p=0.301$, $\eta^2=0.017$	$F(1,64)=0.00$, $p=0.997$, $\eta^2=0.00$	0.462	0.465
TBI	0.918 (0.022)	1.00 (0.02)	0.95 (0.02)	0.99 (0.02)	$F(1,64)=2.01$, $p=0.161$, $\eta^2=0.031$	$F(1,64)=4.30$, $p=0.042$, $\eta^2=0.07$	0.016	0.645
cCBIF	5.87 (0.18)	4.95 (0.18)	5.90 (0.17)	5.22 (0.17)	$F(1,64)=1.74$, $p=0.192$, $\eta^2=0.030$	$F(1,64)=1.10$, $p=0.298$, $\eta^2=0.019$	0.842	0.102
cE-Stage	1.59 (0.25)	2.79 (0.26)	1.28 (0.25)	2.28 (0.25)	$F(1,64)=5.79$, $p=0.019$, $\eta^2=0.092$	$F(1,64)=0.336$, $p=0.564$, $\eta^2=0.006$	0.198	0.041

Results presented as means (standard errors); effect size presented as eta-squared, following Cohen's (1988) recommendations for 0.01 (low), 0.06 (moderate), and 0.14 (high)

NS non-surgically treated eyes, S surgically treated eyes, ARC anterior radius of curvature, ART Ambrósio relational thickness, Avg average, BAD-D Belin–Ambrósio deviation index, BCVA best-corrected visual acuity, CBI Corvis biomechanical index, cCBIF Corvis biomechanical factor, cE-Stage Corvis biomechanical E-staging, CKI central keratoconus index, DA-ratio deformation amplitude ratio, D diopter, IHA index of height asymmetry, IHD index of height decentration, IR integrated radius, ISV index of surface variance, IVA index of vertical asymmetry, KI keratoconus index, Kmax maximum keratometry, Kmean mean keratometry, Max maximum, PPI pachymetric progression index, PRC posterior radius of curvature, SP-A1 stiffness parameter at first applanation, SSI stress–strain index, TBI tomographic and biomechanical index, TCT thinnest corneal thickness, UCVA uncorrected visual acuity

$F=12.51$, $p=0.001$, $\eta^2=0.17$) without interaction, indicating similar modest thinning in both groups.

The BAD-D showed no overall time effect ($F=0.35$, $p=0.559$, $\eta^2=0.01$) but a significant interaction ($F=8.76$, $p=0.004$, $\eta^2=0.12$), with Sidak tests indicating improvement in ICRS eyes and stability in non-operated eyes.

Corneal Biomechanics

DA-ratio, IR, SP-A1, CBI, and SSI showed no significant time or interaction effects, indicating biomechanical stabilization in both eyes. Representative multimodal tomography and biomechanics analyses from a

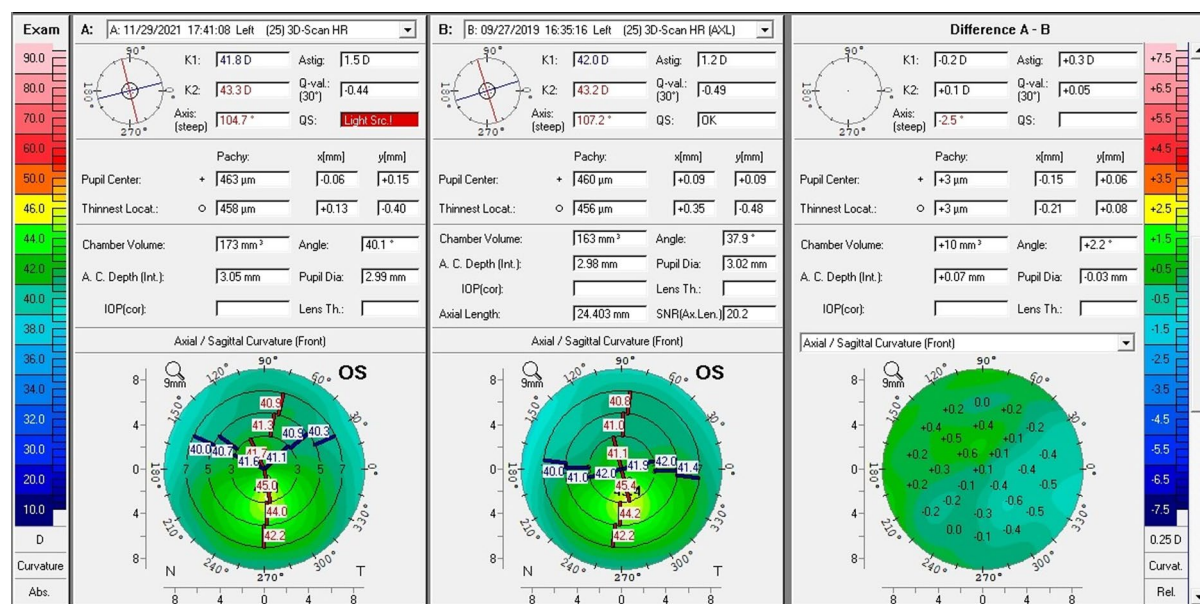


Fig. 1 Anterior axial curvature maps (Pentacam) of the non-operated left eye. Baseline examination performed before contralateral ICRS implantation (September 2019) (A) and the most recent follow-up (November 2021) (B)

are shown, with the corresponding topographic comparison (difference A – B, right). The comparison demonstrates stable anterior curvature over a 26-month follow-up period. *ICRS* intracorneal ring segment

non-operated eye without evidence of progression are shown in Figs. 1 and 2. In contrast, the TBI showed no main effect of time ($F=2.01$, $p=0.161$, $\eta^2=0.031$) but a significant interaction ($F=4.30$, $p=0.042$, $\eta^2=0.07$). Sidak comparisons indicated stability in ICRS eyes ($1.00 \pm 0.02 \rightarrow 0.99 \pm 0.02$; $p=0.645$) and a small increase in non-operated eyes ($0.918 \pm 0.022 \rightarrow 0.950 \pm 0.020$; $p=0.016$). cCBIF showed no significant effects. cE-Stage showed a time effect ($F=5.79$, $p=0.019$, $\eta^2=0.09$) without interaction; Sidak post hoc indicated that the shift was driven predominantly by ICRS eyes.

Keratoconus Progression

Disease progression was observed in two non-operated eyes (6.1%) during follow-up (Fig. 3), both of which were subsequently referred for CXL.

DISCUSSION

In this paired-eye study of patients with clinically asymmetric keratoconus who underwent unilateral ICRS implantation, the non-operated fellow eyes remained stable over a median follow-up of 31 months. Across visual, tomographic, and biomechanical domains, no clinically meaningful deterioration was detected. The only statistically significant change—a small rise in the TBI ($\Delta \approx 0.03$)—occurred without concordant changes in keratometry, pachymetry, tomographic or biomechanical parameters, indicating no structural progression. Importantly, CBI and TBI are designed as quasi-binary discriminators for ectasia detection rather than severity or progression metrics. Their outputs essentially distinguish healthy from keratoconic corneas, but they were not intended to grade disease stage. Severity assessment is better captured by the biomechanical E-staging (cE-Stage) framework,

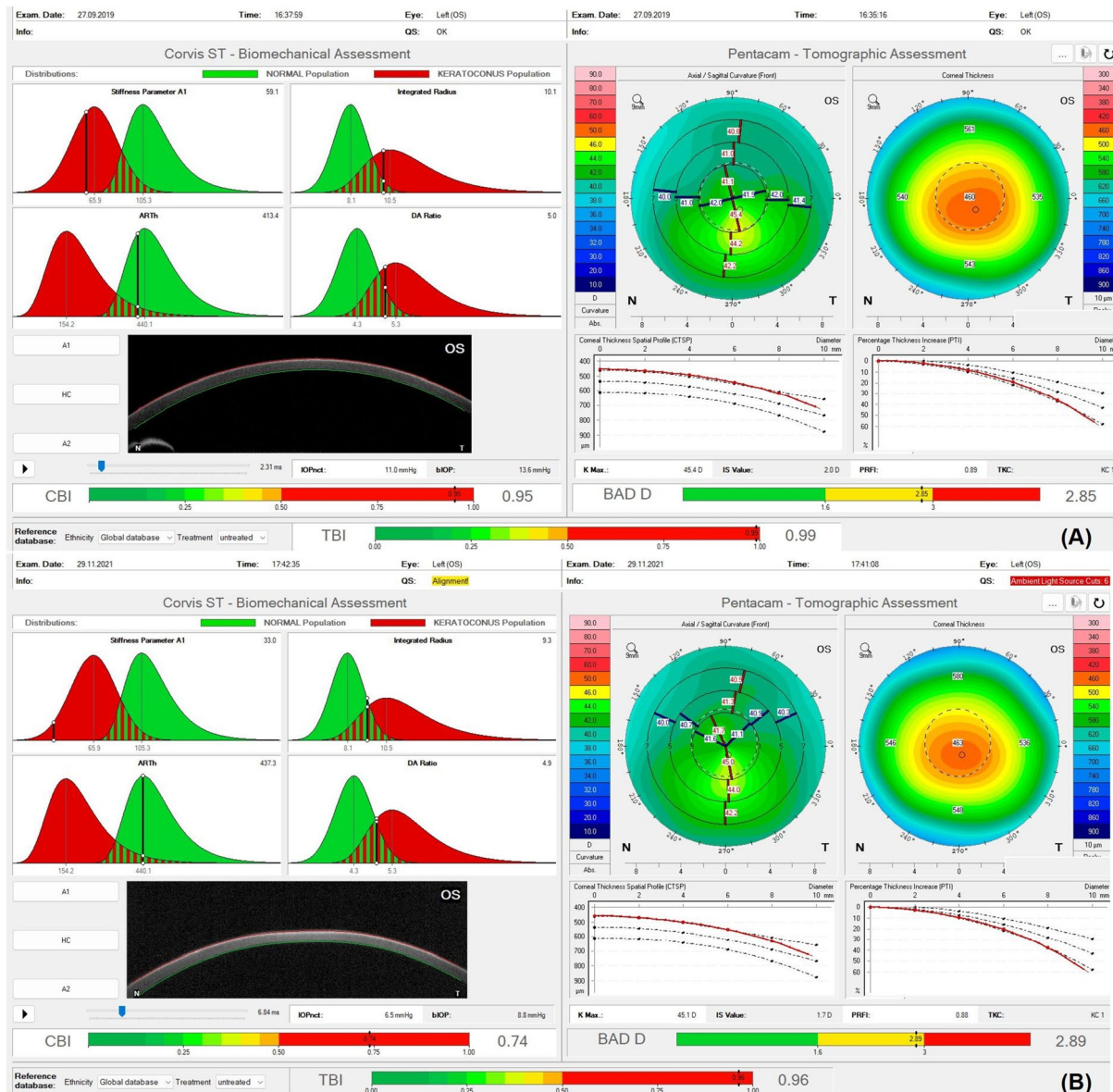


Fig. 2 Pentacam® and Corvis ST® Ambrósio–Roberts–Vinciguerra display of the non-operated left eye obtained before contralateral ICRS implantation (A) and after a 26-month follow-up period (B). The follow-up examination demonstrates overall biomechanical stability, reflected by an increase in ART and reductions in IR, DA-Ratio, and CBI, with only a slight decrease in SP-A1. BAD-D

remained stable, indicating tomographic stability throughout the follow-up period. ART Ambrósio relational thickness, BAD-D Belin–Ambrósio deviation index, CBI Corvis biomechanical index, DA deformation amplitude, ICRS intracorneal ring segment, IR integrated radius, SP-A1 stiffness parameter at first appplanation

which provides a continuous scale suitable for monitoring temporal changes in corneal biomechanics [28]. Therefore, the slight TBI fluctuation we observed should not be interpreted as disease worsening when unaccompanied by

changes in established tomographic or biomechanical parameters.

Overall, the disease course of the non-operated eyes remained essentially stationary under conservative, behavior-based management.

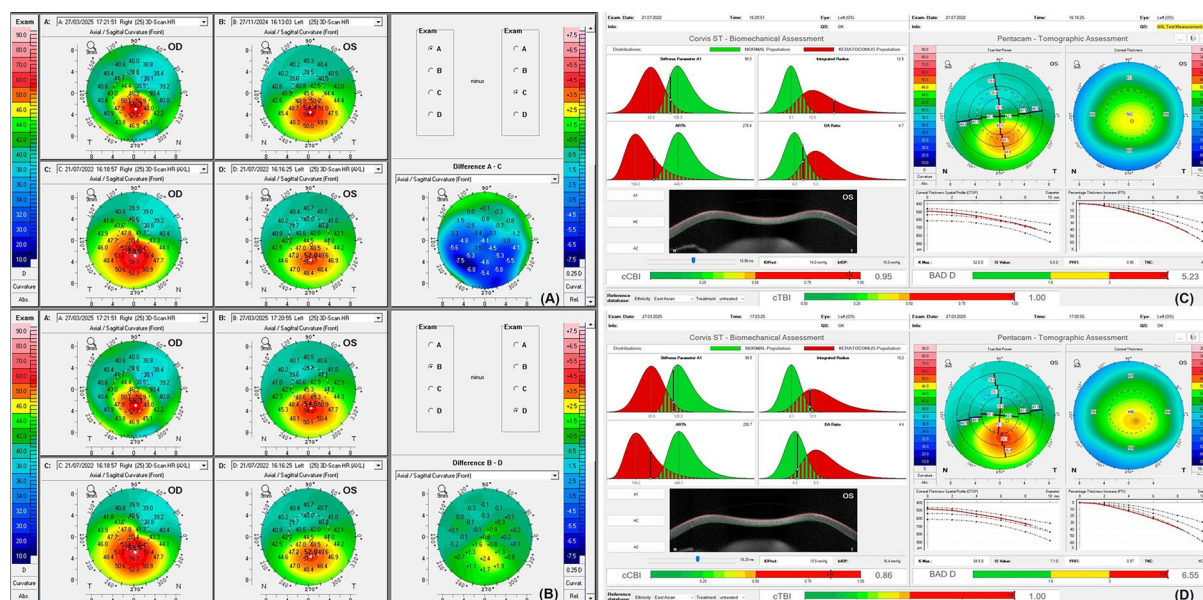


Fig. 3 Multimodal assessment of a patient with tomographic progression in the non-operated fellow eye. Left: A Anterior corneal curvature maps showing marked flattening of the right eye after ICRS implantation (difference A - C), and B an increase in anterior curvature in the non-operated left eye over the same period (difference B - D).

In contrast, the operated eyes exhibited the expected optical improvements following ICRS implantation, with gains in UCVA/BCVA and enhanced corneal regularity.

Baseline Profile of Fellow Eyes

At inclusion, fellow eyes were predominantly mild to moderate ($K_{max} 48.3 \pm 0.8$ D; TCT 488 ± 9 μ m; BAD-D 4.11 ± 0.46 ; cE-Stage 1.59 ± 0.25) with good BCVA (0.09 ± 0.04 logMAR) and biomechanical values typical of early ectasia (SP-A1 77.5 ± 2.6 ; SSI 0.85 ± 0.03). These values positioned the fellow eyes below standard thresholds for CXL or ICRS, justifying observation.

Role of Behavioral Management

A central feature of our cohort was routine conservative care, including patient education on keratoconus, explicit counseling to avoid eye

Right: Pentacam and Corvis Ambrósio–Roberts–Vinciguerra tomographic-biomechanical assessment of the left eye demonstrates an increase in BAD-D from 5.23 (C) to 6.55 (D), consistent with tomographic progression. ICRS intracorneal ring segment

rubbing, oral riboflavin supplementation, and management of allergic eye disease/atopy as clinically indicated. Because this was a retrospective real-world study, adherence to these recommendations was not formally quantified and could not be incorporated as an analytical variable. Mounting evidence links eye rubbing to the onset, asymmetry, and progression of keratoconus, and suggests that cessation may reduce progression risk in a meaningful proportion of cases [9–11]. In this context, the low fellow-eye progression rate observed in our cohort is concordant with prior reports, while not allowing causal attribution to behavioral measures alone.

Comparison with Prior ICRS Literature

The operated eyes in our series exhibited improved acuity and reduced irregularity indices, consistent with long-term studies showing sustained cone flattening and optical quality gains after ICRS [20–24, 26–28]. Importantly,

ICRS mainly redistributes stromal lamellae and regularizes corneal shape; it is not a stiffening procedure per se, which explains why substantial shifts in biomechanical parameters are often absent despite improved optics [19–24]. Our results fit this mechanistic profile.

Progression Criteria and Event Rate

Progression thresholds were defined a priori using multimetric criteria, reflecting contemporary guidance and registry analyses aimed at reducing false positives from single-metric noise [29, 30]. Under these criteria, only two fellow eyes ($\approx 6\%$) progressed and were cross-linked, a rate consistent with carefully monitored, sub-threshold cohorts in the literature [28–30].

Clinical Implications

The present findings have several clinical implications. First, in carefully selected eyes that do not yet meet tomographic or biomechanical criteria for intervention, conservative management with counseling to avoid eye rubbing and control of allergic eye disease, combined with oral riboflavin supplementation and ongoing surveillance, was associated with stability in most fellow eyes. Second, progression should be assessed using multimodal and quantitative criteria rather than relying on isolated keratometric changes. Combining structural parameters such as Kmax, TCT, and BAD-D with thickness-profile indices such as ART or PPI, and biomechanical measures such as SSI or cE-Stage provides a more comprehensive assessment of corneal behavior and minimizes false-positive progression due to measurement variability or epithelial remodeling.

Taken together, these results highlight that individualized monitoring combined with conservative management may help maintain stability in many fellow eyes with asymmetric keratoconus, reserving surgical interventions for objectively progressing cases.

A crucial practical implication of our findings is that behavior-first management cannot be dissociated from systematic follow-up. The stability observed in most fellow eyes occurred

in a setting in which patients were repeatedly educated, monitored, and reassessed with multimodal imaging over a median of 31 months.

The detection of progression in 6% of fellow eyes further reinforces that, even when initial tomographic and biomechanical metrics are subthreshold for CXL or ICRS, clinicians should maintain regular, protocol-driven surveillance. In our view, conservative management of asymmetric keratoconus should be framed as an active, follow-up-dependent strategy that combines behavioral change and scheduled multimodal examinations, rather than as “no treatment”.

Strengths and Limitations

Strengths of this study include its paired-eye design, which provides an intrinsic control for interindividual variability, and the use of standardized tomographic and biomechanical assessments obtained with the same devices throughout follow-up. This multimodal and internally controlled design strengthens the validity of the longitudinal comparisons. Limitations include the retrospective nature of the study, a non-protocolized follow-up schedule, potential undercapture of adherence to behavioral measures, and multiplicity across endpoints (mitigated by the Sidak correction and emphasis on effect sizes). The two fellow-eye CXL crossovers were kept in the intention-to-observe analysis, which is conservative with respect to stability estimates.

The low number of progression events in fellow eyes (two eyes, 6.1%) is clinically reassuring but limited the ability to perform statistically robust subgroup or multivariable analyses to identify baseline predictors of progression, such as age or tomographic and biomechanical severity. Accordingly, while this study was not designed to define prognostic risk factors, it highlights the need for larger prospective or registry-based cohorts to further refine risk stratification.

Prospective work that quantifies behavioral adherence and anchors progression thresholds to repeatability limits will further refine patient selection and follow-up intervals. Although the

median follow-up exceeded 2.5 years, with some eyes followed for more than a decade, a subset of patients contributed only the minimum 12-month interval, and follow-up was anchored to the last available visit meeting the predefined imaging requirements. It is therefore possible that additional progression events may emerge beyond the time frame captured in this dataset. Our results should not be interpreted as evidence that further follow-up is unnecessary, but rather as support for the notion that, within the limits of the present study, a behavior-first strategy combined with ongoing surveillance is safe for carefully selected fellow eyes. Prospective studies with standardized, longer-term follow-up are needed to refine prognostic risk stratification.

CONCLUSIONS

In this paired-eye cohort of clinically asymmetric keratoconus treated unilaterally with ICRS, the non-operated fellow eyes remained stable over a median follow-up of 31 months, showing no clinically significant changes across visual, tomographic, or biomechanical domains. Most fellow eyes maintained stability under conservative management, including counseling to avoid eye rubbing, oral riboflavin supplementation, and management of ocular allergy/atopic disease as clinically indicated, while the operated eyes demonstrated the expected optical improvements following ICRS implantation.

However, disease progression was observed in two eyes (approximately 6%), which were subsequently referred for corneal cross-linking. These findings support a conservative, behavior-first approach combined with oral riboflavin and active multimodal surveillance incorporating both tomography and biomechanics, while reserving interventional treatment for eyes demonstrating objective progression. Given the real-world retrospective design and the lack of formal quantification of adherence, stability should not be attributed solely to behavioral measures; rather, our results suggest that carefully selected fellow eyes at subthreshold levels can be managed conservatively, provided regular follow-up is maintained. In this context,

integrating regular, protocol-driven multimodal assessments is key to ensuring that only cases with actual disease progression receive more invasive treatments, thereby optimizing patient care while minimizing the risk of missing late-onset deterioration.

Author Contributions. João A. Ambrósio, Júlio Brissos, and Renato Ambrósio Jr. conceptualized and designed the study. João A. Ambrósio, Júlio Brissos, Gonçalo C. Tardão, Alexandre Neto, Louise Esporcatte, and Renato Ambrósio Jr. prepared the material, collected the data, and analyzed the data. João A. Ambrósio and Júlio Brissos wrote the first draft of the manuscript, and all authors commented on and approved the final version.

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Data Availability. The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of Interest. Prof. Renato Ambrósio Jr. is a consultant for Oculus, Alcon, Zeiss, and Mediphacos. Renato Ambrósio Jr. is an Editorial Board member of *Ophthalmology and Therapy*. Renato Ambrósio Jr. was not involved in the selection of peer reviewers for the manuscript nor any of the subsequent editorial decisions. João A. Ambrósio, Júlio Brissos, Gonçalo C. Tardão, Alexandre Neto, and Louise Esporcatte have nothing to disclose.

Ethical Approval. This study, conducted at the Instituto de Olhos Renato Ambrósio, was approved by the Comitê de Ética em Pesquisa da UNIFESP—Universidade Federal de São Paulo—Hospital São Paulo (approval number 4.050.934; 27784919.0.0000.5505). All procedures were conducted in accordance with the ethical standards of the institutional and national research committees and with the

principles of the Helsinki Declaration of 1964 and its later amendments. Given the retrospective design, informed consent was waived, and only anonymized, non-identifiable clinical data were used in compliance with institutional guidelines and data protection standards.

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